Thomas Biju Kavukatt

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Open to relocation anywhere in India · Notice period 2 months (negotiable)

SOFTWARE ENGINEER - C++ DESKTOP APPLICATIONS | GTKmm · Qt · GStreamer · Linux

One year at Digital Core Technologies building the desktop software for thermal imaging cameras in C++ (GTKmm, now moving to Qt): Ethernet streaming, on-device recording and storage, and a motorised zoom-lens controller. Comfortable with the parts most UI developers avoid: GStreamer pipelines, camera streaming, and debugging a codebase customers already run. Build full-stack web and mobile apps on my own time. B.Tech CSE, 2025.

TECHNICAL SKILLS

Languages: C++, Python, TypeScript / JavaScript, SQL, Java

Desktop / UI: GTKmm, Glade, Qt (UI porting work), event-driven GUI design

Video & imaging: GStreamer pipelines, RTSP/H.264 and YUV422 streaming, on-device recording and storage, real-time frame handling, bad-pixel correction, zoom/focus control for motorised optics

Web & mobile: React, React Native / Expo, Node.js, Supabase (Postgres, Auth, Row Level Security, Realtime), Tailwind

Tools: Git, Linux, Vercel, Android APK builds

PROFESSIONAL EXPERIENCE

Digital Core Technologies, Kochi *Jul 2025 – Present*
Software Engineer (Oct 2025 – Present) · Software **Engineering Intern** (Jul - Sep 2025, converted to full-time)
ThermView - desktop control and viewing software for thermal imaging cameras. C++, GTKmm, GStreamer, Linux.

- Built the streaming and storage features for the Ethernet thermal camera: recording and snapshots to on-device storage, storage format and stream bitrate control, from the GTKmm UI through the device SDK.
- Added H.264/YUV422 stream format selection with the GStreamer source pipeline rebuilt on the fly; fixed frame drops by sourcing the RTSP feed once and sharing it across pipelines.
- Hardened recording against reboot, power-off, factory reset and format changes mid-recording.
- Built the zoom and focus control front end for a motorised optical zoom-lens camera, wired over the device link.
- Added bad-pixel correction to the live view: the operator picks defective pixels with the mouse on the running image and the fix applies immediately.
- Ported UI components as part of the ongoing move of ThermView from GTKmm to Qt.
- Cut ThermView releases, integrated SDK updates and reviewed code on the shipped codebase; regressions are mine to chase.

Accel Limited, Kochi - Software Testing Intern *Jul 2024*
Functional testing on a client application. Wrote up and tracked bugs with the dev team.

PROJECTS

ChoreWars - shared-house cleaning rotations with a weekly league *2026*
React 19, TypeScript, Vite, Tailwind, Supabase (Postgres, Auth, Realtime), PWA

- Live and used every day by the six people in my house: who cleaned what last, who is next, what is due, and a points league on top.
- All scoring rules (rotation advancement, volunteering, streaks, cooldowns, achievements) live in Postgres functions. The browser only calls RPCs like complete_cleaning(area_id) and never sends a points value, so nobody can farm points from the client.

CardWallet - Android card wallet app *2025*
React Native, Expo, TypeScript, Supabase (Postgres, Auth, RLS)

- Stores debit, credit, ID, membership and business cards, each with its own card-type layout. Built solo, runs on my phone as an APK.
- Card details stay hidden until the user passes fingerprint/face authentication.

VeinChain — finger-vein biometric authentication (final-year project) *2025*
Python, ResNet-50, near-infrared imaging, blockchain

- Built the NIR capture rig: removed the IR-cut filter from a webcam and lit the finger with NIR LEDs from behind, so the veins absorb the light and show up dark in the image. Wrote the capture pipeline and the front end.
- A ResNet-50 model matches the vein pattern; authentication results are written to a blockchain ledger for tamper-evidence. Team project.

EDUCATION

B.Tech, Computer Science & Engineering - Rajagiri School of Engineering & Technology, Kochi · CGPA 7.88 *2021 – 2025*

Class XII (CBSE) - Devagiri CMI Public School, Kozhikode · 95.6% *2021*